Harish Balaji V

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Professional Summary

AI engineer specialising in LLM system design, RAG architecture, and production evaluation pipelines. Built and deployed a multi-document RAG chatbot (LangChain, chroma, 500+ docs) and a 7-dimension LLM evaluator targeting >90 perecent quality scores at WABTEC. Designed LLM-powered QA automation at Indium Software that improved test coverage by 30 percent. Published in Routledge (2025) on agentic AI integration with ML/DL systems. Graduated in May 2026 — available for immediate full-time joining.

Experience

ML & Gen AI Engineer Intern **Jun 2025 – May 2026**
WABTEC Industrial India Pvt Ltd · Remote Monitoring and Diagnosis

- Built a robust RAG pipeline using LangChain, FAISS vector store, and a hosted LLM over 300–500 service manual documents, accelerating field mechanic document retrieval speed by 35% compared to manual search methods.
- Designed an LLM-based evaluation framework with specialized prompt engineering across 7 quality dimensions (accuracy, hallucination, coherence, relevance, safety, completeness, and consistency) to secure a composite quality target of >90%.
- Improved component failure classification accuracy by engineering ML-ready datasets from raw operational logs using Pandas, NumPy, and Scikit-learn, cutting manual data preparation effort by ∼40% via automated preprocessing.
- Reduced unplanned downtime risk across 3+ critical failure categories by developing and optimizing KNN and ANN predictive models integrated with intelligent component labeling and advanced feature engineering.
- Surfaced deep mechanical wear anomalies in lubrication behavior by conducting multi-parameter Exploratory Data Analysis (EDA) with Matplotlib and Seaborn, directly optimizing maintenance scheduling decisions for fleet managers.

AI Engineer Intern – Quality Engineering Division **Dec 2024 – Mar 2025**
Indium Software (India) Pvt Ltd

- Improved test coverage by 30% by designing LLM-powered test case strategies leveraging natural language understanding to generate dynamic, human-like input variations, reducing manual test authoring cycles.
- Accelerated bug triage turnaround by contributing to an intelligent bug-tracking pipeline using LLM-based semantic similarity and anomaly scoring, enabling autonomous agents to surface duplicates early in the CI/CD workflow.
- Strengthened validation coverage for AI-driven QA frameworks by integrating autonomous agents with LLM backends for test automation and error detection, supporting reliable deployment of NLP-based testing at scale.

Technical Skills

LLM & Gen AI: LangChain · LangGraph · LlamaIndex · RAG pipelines · FAISS · Chroma · Prompt Engineering · LLM Evaluation · Fine-tuning· HuggingFace Transformers · OpenAI API · Azure OpenAI · AWS Bedrock
ML / DL: PyTorch · TensorFlow / Keras · Scikit-learn · XGBoost · SHAP · Computer Vision · NLP · Reinforcement Learning
MLOps & Cloud: FastAPI · Docker · MLflow · CI/CD · AWS (SageMaker, S3, EC2) · Azure ML · GCP

Languages & Data: Python · SQL · Pandas · NumPy · Matplotlib · Seaborn

Projects

Agentic System for Assessing Water Quality Along Indian Rivers **Jan 2026 – May 2026**

- Built a 5-agent autonomous pipeline (Data, Preprocessing, ModelTrainer, Evaluator, and LLM Reasoning) for end-to-end Water Quality Index (WQI) prediction on CPCB river datasets.
- Analyzed 7 physicochemical parameters (DO, pH, BOD, etc.) to train ML regression models, achieving optimal selection via R^2, MAE, and RMSE benchmarks.
- Applied SHAP-based explainability to identify key predictive features and enhance model interpretability.
- Integrated LangChain with OpenAI/Gemini to generate human-readable reports and actionable recommendations.

Agent AI For Precision Agriculture: Integrated Approach to Crop Management **Jan 2025 – May 2025**

- Developed an agentic AI framework integrating real-time climate, pesticide, and production data for crop yield prediction and recommendation across 18 crop types.
- Built a dual-model architecture: Random Forest Regressor for yield prediction and a Deep Neural Network (Keras/TensorFlow) for suitability classification with early stopping optimization.
- Resolved "black-box" interpretability issues by designing an interface with LLM-powered explanations , making complex model decisions accessible to non-technical users.
- Integrated an Agentic AI decision engine to autonomously route inputs, aggregate ML/DL outputs, and generate science-backed farming recommendations.
- *Published as Chapter 52 in "Intelligent and Sustainable Systems" (Routledge, 2025), ISBN: 9781041302919.*

Certifications

- Generative AI: Introduction and Applications, IBM(Coursera) – 2024
- Generative AI: Prompt Engineering, IBM (Coursera) – 2024
- Inferential Statistics, University of Amsterdam (Coursera) – 2024
- Cloud Foundations, Amazon Web Services – 2023
- Machine Learning Foundations, Amazon Web Services – 2023
- Python Basic, HackerRank – 2023
- SQL Intermediate, HackerRank – 2023
- NoSQL Databases Essentials, IBM (Coursera) – 2023

Education

Integrated MTech in Artificial Intelligence **Aug 2021 – May 2026**
SRM Institute of Science and Technology, Kattankulathur
GPA: 8.6/10
Coursework: Machine Learning, Inferential Statistics and Predictive Analysis, Neural Networks, Deep Learning, Software Engineering and Project Management, Natural Language Processing, Data Structures and Algorithms, Database Management Systems, Accelerated Data Science

12th Grade - Physics , Chemistry , Computer Science **Jul 2021**
Bharathi Vidyalaya Senior Secondary School (Percentile: 83)

10th Grade **May 2019**
Bharathi Vidyalaya Senior Secondary School (Percentile: 82.7)